RAPHAEL ESQUIVEL

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EDUCATION

Harvey Mudd College, Claremont, CA — Aug 2025 - May 2029
B.S., Mathematics & Physics — GPA: 3.88 (Major: 3.93)
Relevant coursework: Electromagnetic Fields, General Relativity, Quantum Field Theory, Nanomaterials, Engineering Systems, Abstract Algebra I, Intermediate Probability, Numerical Analysis, Readings in Algebraic Geometry

Arizona State University, Phoenix, AZ — Aug 2024 - Apr 2025
Non-Degree Seeking Student — GPA: 4.0
Relevant coursework: Mathematical Methods Physics II, Introductory Modern Physics, Quantum Physics II, Classical Particles, Fields, and Matter II

EXPERIENCE

Research Assistant, Quantum Quasars Lab — Sep 2025 - Present
Advisor: Prof. Jason Gallicchio (Harvey Mudd)

- Designing and assembling a free-space quantum communication system for a kilometer-scale demonstration of quantum position verification.
- Developed an ultra-low-latency RF synchronization link and FPGA-based control for RF communication and photonics.
- Built and aligned optical experimental setup and fabricated custom optomechanical components.

Student Researcher, Logarithmic Gromov–Witten Theory — Nov 2025 - May 2026
Advisor: Prof. Dagan Karp (Harvey Mudd)

- Analyzed the tropical and combinatorial structure of relative stable maps to $(\mathbb{P}^n, H)$ and their boundary behavior in moduli spaces.

Student Researcher, Geometric Mechanics — Aug 2024 - May 2025
Mentor: Prof. Sarah Post (University of Hawai'i at Mānoa)

- Studied Lie algebroids, including Atiyah algebroids of principal bundles, in constrained Lagrangian systems, relating their geometry to the Boltzmann–Hamel equations.

Repair Steward, HMC Makerspace — Aug 2026 - Present

AWARDS

Karl Menger Memorial Award (2nd), American Mathematical Society — May 2025
Awarded $1000 at Regeneron ISEF for project *Lie Groups and Algebroids in Lagrangian Reduction.*

SKILLS

Programming languages	Python, Javascript, C/C++, R, Mathematica, Julia, Verilog
Software	COMSOL, Xilinx Vivado, Git, Jupyter
Lab equipment	Free-space optics, lasers, SPAD, Pockels cells, optical fiber, AFM, SEM/STEM, Raman and UV–Vis spectroscopy, XRD
Manufacturing	FDM, SLS, lathe, mill, 3D printer maintenance
Languages	English (native), French (native), Spanish (proficient)

RESEARCH INTERESTS

Mathematical physics, quantum field theory, algebraic geometry, and quantum information.